

Evaluating Emoji Semantics: A Comparison of Computational Embeddings and Human Valence Judgements

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ABSTRACT

Emoji embedding models are widely used to represent emoji meaning in natural language processing tasks, yet their alignment with human emoji interpretations remains unexplored. We evaluated the validity of emoji embeddings by examining whether computational representations of emoji could correspond to human judgements of emotional valence. We compared two embedding models, emoji2vec and emojiional, using the Emoji Sentiment Ranking dataset derived from large-scale human-annotated data as ground truth. Emoji embedding vectors were extracted and used as input features for classifiers to predict emoji sentiment polarity (i.e., positive, negative, or neutral). Model performance was evaluated using accuracy and F1 scores, which balance between precision and sensitivity. To address the issue of class imbalance, we applied Synthetic Minority Oversampling and downsampling to balance the samples in each class. Our results showed that vector embeddings of emoji do seem to preserve emotional valence. Support Vector Machines achieved a macro F1 of 64%, almost twice as high as the naive baseline. However, best results were only achieved after balancing class sizes within the dataset. This work displays the extent to which emoji embedding space preserves human valence judgements and how training data can affect the effectiveness of emoji embeddings in capturing human-like valence meanings.

1 Introduction

Since their creation in 1999, emoji have become ubiquitous in digital communication. In the context of texting and social media, emoji are used as additional emphasis, replacements for words, and tone indicators. Due to the visual nature of emoji, they are processed differently in the brain than words. Depending on their clarity and role in the text, emoji take more time and effort to process [1,2]. In some cases emoji semantics and pragmatics can even be more volatile than written language. Similar to words, a single emoji can be interpreted differently depending on a variety of factors. For instance, concreteness, root word meaning, and visual design can alter how one derives meanings in emoji [3].

Broader contexts also affect how emoji are used and understood. Individual differences between both authors and audiences, current events, cultural contexts, changes over time, language differences, and even the device that one views the emoji on can alter emoji semantics. Thus, linguists researching digital communication including emoji and Natural Language Processing (NLP) research on emoji meanings must be able to account for the contexts in which they appear and to compare against human interpretations of emoji meanings..

2 Related Work

NLP research on linguistic data involves using computational techniques to mathematically process language. One such technique is creating vector embeddings, which are mathematical representations of linguistic inputs in multi-dimensional vector space that quantify semantic information. One of the first attempts in creating vector embeddings is the construction of word2vec [4], where a word's meaning is evaluated in a large corpus of text. Specifically, the model learned semantics of each word based on its co-occurrence with other words. This approach quantifies meanings of each word within a 300-dimensional vector space. The position of a word is defined by a vector, which is learned based on its closest neighbors in the text during the training of the word2vec model. Embedding vectors naturally preserve semantic relationships, especially between commonly co-occurring words.

Building on this technique, researchers studying emoji have developed vector embeddings that include emoji. One of the most widely used emoji embeddings is emoji2vec [5]. Emoji2vec represents emoji in a 300-dimensional vector space that is compatible with the word2vec embeddings. Despite its popularity, the validation of emoji2vec has been criticized, and several studies have shown that alternative emoji embedding models outperform it in capturing semantic information. These models include, but are not limited to, emojipace [6], DeepMoji [7], and emojiional [8]. Unlike emoji2vec, these models are trained on substantially larger datasets containing emoji in naturalistic contexts and employ more sophisticated modeling techniques, such as attention mechanisms, to learn emoji representations.

To directly compare newer models against the classic emoji2vec model, it would be more appropriate to select a model that was built using similar machine learning techniques. This points to emojiional as both emoji2vec and emojiional are based on word2vec and are constructed from large collections of English tweets. Emoji2vec was created in 2016 and contains 1661 emoji. In contrast, emojiional was created in 2021, containing a larger number of emoji (1816). In addition, emojiional was built from a larger and more comprehensive corpus than emoji2vec.

To validate mathematical analyses in linguistics, researchers must understand whether or not their computational algorithms accurately reflect and preserve human understanding and judgment of meaning. One facet of semantics that vector embeddings should be able to preserve is Emotional Valence (EV), also known as sentiment polarity. EV refers to whether a paragraph, sentence, or word evokes a positive, neutral, or

negative reaction [9]. While there is very little work on emoji EV, studies predicting word EV with machine learning classifiers have found mixed results. Rexha et al. [10] created an algorithm to classify whole tweets as positive or negative, finding poor results. Although they found F1 rates above 90% for positive tweets, they failed to classify negative tweets accurately, only reaching a 54% F1 score at maximum. The authors noted that both the training and testing datasets used were heavily imbalanced, with each being composed of around 80% positive entries. Other studies have found similar levels of success as well as similar limitations. Kashyap et al. [11] assessed a variety of models' performance analyzing several embeddings, but only reached a Macro F1 of 69%, indicating moderate success.

Indeed, data imbalance is a notable issue in machine learning studies, as heavily imbalanced datasets can cause problems for training and testing. Previous literature has shown that EV datasets tend to include more positive data, leading to a natural imbalance. There are several ways to address imbalanced data. For instance, Xu et al. [12] assessed a new method of oversampling, which is a method to generate additional data from a minority class in order to ameliorate class imbalances while avoiding overfitting risks. They generated specific semantic data, then applied the Synthetic Minority Oversampling Technique (SMOTE), finding that its use on their variables improved model performance and did not cause additional issues for classification.

While vector embeddings incorporating emoji have been used in computational semantic research, little work has examined their ability to preserve EV, especially in English. The majority of work has examined using emoji to improve EV analysis of the text in which the emoji appears, instead of directly predicting the valence of the emoji itself [13]. As such, there is a gap in the literature concerning the validity and quality of emoji vector embeddings' ability to preserve emotional valence, and other related semantic information.

3 Objectives

The objective of this study was to evaluate whether vector embeddings preserve emoji EV. To this end, we trained a suite of machine learning models on two separate emoji vector embeddings (emoji2vec [5] and emojiional [8]) to predict the EV of individual emoji, as judged by a panel of humans. Models trained included XGBoost, Support Vector Machines (SVM), and Random Forest models. All three models have been used in similar research, with XGBoost specifically known to perform well with imbalanced classes, and SVM are useful with high-dimensional data. Although the vector embeddings are similar to each other, the emoji2vec embedding was created closer in time to the set of human judgments, and would therefore be at less risk of semantic drift due to time. As a result, we theorized that:

Hypothesis 1: Machine learning classifiers will predict emoji EV accurately, at a rate above baselines.

Hypothesis 2: Of the classifiers, XGBoost will perform best, SVM will perform second best, and Random Forest will perform the worst.

Hypothesis 3: Classifiers trained on emoji2vec embedding vectors will yield better performance on their predictions than

emojiional.

4 Methodology

This section first describes the embedding spaces and datasets involved in the project, followed by two methodological steps. First, we extracted embedding vectors of emoji from our embedding spaces. Second, we trained three classifiers to evaluate performance of the emoji embedding models on preserving human emotional valence judgements.

4.1 Embedding Space and Datasets

Due to constraints related to model availability, our analyses focused on two embedding models: emoji2vec and emojiional. Emoji2vec [5] is an emoji embedding model that maps emoji into the same semantic space as the pre-trained word2vec embeddings. The emoji2vec model constructed emoji vectors by summing the word2vec embedding vectors of descriptions around Unicode emoji (i.e., description vector) and defining a trainable vector for each emoji. Then, the model was trained to maximize the cosine similarity between an emoji's trainable vector and its corresponding description vector. This process resulted in 300-dimensional representations of emoji that aligned with words in the word2vec space. Therefore, using emoji2vec embedding, one can directly compare the meanings of emoji and words.

Building on emoji2vec, emojiional [8] is designed to capture a wider range of affective and sentiment properties of emoji meaning that are not constrained to categories like positive or negative. Instead of using the official Unicode descriptions, they collected keywords from online emoji dictionaries including Emojipedia and emoji.wiki. This allowed them to capture more multifaceted meanings that emoji2vec may miss. They trained the embeddings by tasking the model to predict the specific emoji given a set of input keyword vectors. This approach allowed a single emoji to represent a range of emotional content rather than being constrained to a single fixed definition. Like emoji2vec, the resulting Emojiional embedding space is also a pre-trained set of embeddings where emoji and words exist in the same 300-dimensional space.

To evaluate the embeddings against human judgements of sentiment, we used the Emoji Sentiment Ranking dataset [6]. This dataset contains sentiment scores for 751 frequently used emoji derived from over 1.6 million human annotated tweets where each tweet was labelled as negative, neutral, or positive. Other key fields include the emoji's occurrence count, the emoji's average position within a tweet, and the proportion of times the emoji appears in negative, neutral, and positive tweets.

The final dataset used in this study was composed of the total list of emoji that were included in both the Emoji Sentiment Ranking dataset and the vector embeddings. The resulting dataset contained 300 vectors and the human judgments for 682 emoji. Of the emoji, 433 were considered positive, 182 neutral, and 67 negative. These labels were considered as ground truth of the current project.

4.2 Extracting Embedding Vectors

We used the Python library Gensim to extract embedding vectors of emoji from the selected models, emoji2vec and

emojional. Each emoji is represented as a 300-dimensional vector indexed by its Unicode. Embedding vectors were extracted only for the emoji that are present in the Emoji Sentiment Ranking dataset. Extracted embedding vectors were aligned by emoji Unicode and stored as a separate dataset for analyses.

For emoji2vec, the embeddings were accessed following the instructions provided in its official repository: <https://github.com/uclnlp/emoji2vec>. Emoji2vec maps emoji into the same vector space as word2vec and is provided in a format compatible with standard embedding tools. Emoji vectors were extracted via the Unicode keys in the embedding.

For emojional, the pre-trained embedding file was also accessed from the model's official repository: <https://github.com/elenabarry/emojional/tree/main>. The embedding is distributed in a binary format compatible with word2vec. The model was loaded with the binary option enabled, and each emoji was also represented as a vector accessible through its corresponding Unicode key.

4.3 Classification on Emotional Valence

We trained three classification models to categorize emoji valence based on emoji embedding vectors of emoji2vec and emojional. The models included Support Vector Machine (SVM), Random Forest, and Extreme Gradient Boosting (XGBoost). The models were trained using Python library scikit-learn and XGBoost. These models were trained on the emoji embeddings extracted from emoji2vec and emojional, with the task of sorting the emoji into the "positive", "negative", or "neutral" class. Model parameters were optimized for performance. All models were trained using 10-fold cross-validation on 80% of the data and subsequently evaluated on the remaining 20% holdout set.

SVMs are supervised learning models used for classification and regression tasks. The primary goal of an SVM is to find an optimal hyperplane that best separates data points of different classes in the feature space. We used the Radial Basis Function kernel, which is a nonlinear kernel that effectively handles non-linearly separable data by mapping it to a high-dimensional space. The influence of a single training example on the decision boundary was calculated based on the data (i.e., scaled). Additionally, we used one-versus-one parameters for multi-class classification.

Random Forest is an ensemble learning method that uses Bootstrap Aggregating technique. It constructs a number of decision trees where each tree is trained independently. Output is the majority vote among the trees. Random Forest's strength lies in reducing overfitting and improving generalization due to the averaging of many decorrelated trees. Our number of trees was 100, with a maximum depth of 6. The minimum number of samples required to be at a node was 5. This constraint helps smooth the model and reduce its sensitivity to noise in the data, preventing overfitting.

XGBoost is a gradient boosting algorithm designed to be efficient and flexible. Unlike Random Forest that builds trees independently, XGBoost builds trees sequentially. Each new tree in the ensemble attempts to correct the errors left in the previous trees. It specifically focuses on improving the model's performance on misclassified instances. The model objective was set to output predicted probabilities of each data point belonging

to each class. The number of boating rounds (i.e., trees) was set to 200, with a maximum depth of 6 and a learning rate of 0.1.

Success for each model was evaluated using information in the confusion matrix. Key metrics used in this project included accuracy and F1 score, which is a balance between precision and sensitivity (or recall). Accuracy quantifies the proportion of total predictions that are correct, measuring overall classification performance across all classes. Precision qualifies the proportion of predicted cases that are truly in the corresponding class. It reflects how reliable the positive predictions are per class. Lastly, sensitivity measures the proportion of true cases that are correctly identified. Since our classes were imbalanced, we focused on F1 score as it can better capture performance metrics of precision and sensitivity.

To evaluate statistical significance of model improvements relative to the baseline and to other models, we also conducted non-parametric bootstrapping analyses to estimate 95% confidence intervals for all performance metrics.

5 Analyses and Results

All statistical and modeling procedures were implemented in Python 3.13.7. The experiments consisted of an initial evaluation using the imbalance datasets, followed by the application of class-balancing techniques to assess their impact on model performance.

5.1 Performance on Imbalanced Datasets

The imbalance datasets consisted of 433 positive emoji, 182 neutral emoji, and 67 negative emoji. Model performance was evaluated on accuracy and macro F1, which is the unweighted mean of class-specific F1. Baseline for accuracy was 0.63, macro F1 was 0.333%, F1 (Positive) was 0.63, F1 (Neutral) was 0.27, F1 (Negative) was 0.10. Model performance is shown in Table 1.

Table 1. Classification performance of SVM, Random Forest (RF), and XGBoost (XGB) across emoji2vec and emojional on imbalanced datasets.

	emoji2vec			emojional		
Classifier	SVM	RF	XGB	SVM	RF	XGB
Accuracy	.61	.64	.64	.66	.64	.64
Macro F1	.46	.26	.41	.52	.26	.43
F1 - Pos	.73	.78	.77	.77	.78	.77
F1 - Neu	.25	.00	.20	.34	.01	.23
F1 - Neg	.41	.00	.26	.34	.00	.28

Overall, results showed that accuracy scores were relatively consistent across models, ranging from 0.61 to 0.66. The highest accuracy (0.66) was achieved by SVM with the emojional embedding, while Random Forest and XGBoost showed comparable performance (0.64) across both embedding types. Differences between models became more pronounced when examining macro F1 scores, which better reflect performance under class imbalance. SVM consistently outperformed the other models, achieving 0.46 with emoji2vec and 0.52 with emojional.

In contrast, Random Forest performed poorly in terms of macro F1 (0.26 across both embeddings), indicating difficulty in capturing minority classes. XGBoost showed moderate performance, with macro F1 increasing from 0.41 (emoji2vec) to 0.43 (emojional), suggesting some benefit from the emojional representation.

Class-specific F1 scores further highlight the imbalance issue. All models achieved relatively strong performance on the positive class (0.73 to 0.78), with Random Forest slightly outperforming others (0.78 across both embeddings). However, performance dropped substantially for the neutral and negative classes. This is especially the case for Random Forest, which nearly failed to identify these two classes (0.00 to 0.01). SVM and XGBoost had moderate class-specific F1 scores for neutral and negative classes, but still overly emphasized the positive class.

Statistical comparison of model performance was conducted using bootstrapping analysis. Results for accuracy and macro F1 are shown in Figure 1 and Figure 2, respectively. Results for accuracy suggest no statistical differences among SVM, Random Forest, and XGBoost, and no statistical differences between emoji2vec and emojional. However, while emoji2vec and emojional did not differ in macro F1 scores, SVM and XGBoost performed significantly better than Random Forest. We also observed that Random Forest exhibited nearly absent (for accuracy) or highly skewed (for macro F1) confidence intervals. This indicates minimal variability across bootstrap samples and suggests that the model produced highly consistent predictions. However, this consistency reflects a strong bias rather than robustness. Random Forest for imbalanced dataset systematically favoured certain classes while failing to adequately capture minority classes. Lastly, only SVM and XGBoost (for emojional) achieved significant improvement from the macro F1 baseline.

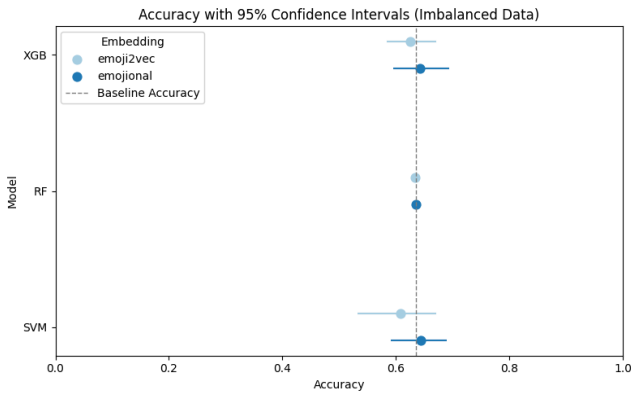


Figure 1. Bootstrapped accuracy of SVM, Random Forest, and XGBoost, across emoji2vec and emojional using imbalanced datasets.

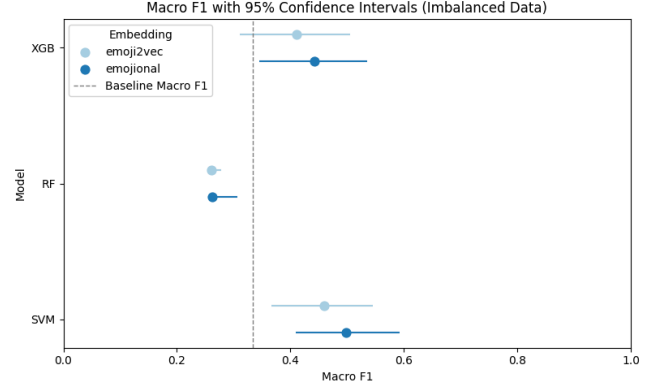


Figure 2. Bootstrapped macro F1 of SVM, Random Forest, and XGBoost, across emoji2vec and emojional using imbalanced datasets.

5.2 Class Balancing

To address class imbalance, we conducted a parametric experiment using the Synthetic Minority Oversampling Technique (SMOTE) combined with downsampling. SMOTE generates synthetic samples for minority classes to improve class representation, while downsampling reduces the size of the majority class to achieve a more balanced distribution. Together, these approaches allow for a controlled evaluation of how class rebalancing impacts model performance. Results are shown in Figure 3.

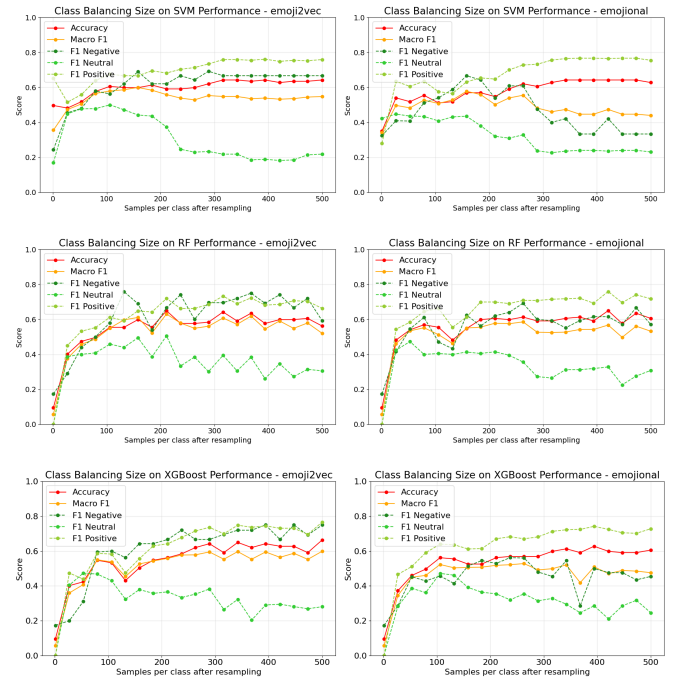


Figure 3. Parametric experiment combining SMOTE and undersampling for SVM, Random Forest (RF), and XGBoost.

Results from the parametric experiment indicated that model performance improved as class balance was introduced, with a

clear trade-off between overall performance and minority class representation. Specifically, increasing the number of samples per class led to gains in overall metrics, but excessive resampling began to degrade performance on the neutral class.

We identified 171 samples per class as an optimal balance point, where overall model performance improved without a substantial loss in F1 score for the neutral class. Beyond this point, gains in aggregate metrics were offset by declines in minority class performance.

5.3 Performance on Balanced Datasets

Based on the findings in the parametric experiment, we adopted a hybrid resampling strategy combining SMOTE and downsampling to standardize all classes (positive, neutral, and negative) to 171 samples each. This approach ensures a balanced dataset while preserving class-specific performance. Baseline for all metrics was .33 for balanced datasets. Model performance is shown in Table 2.

Table 2. Classification performance of SVM, Random Forest (RF), and XGBoost (XGB) across emoji2vec and emojiional on balanced datasets.

Classifier	emoji2vec			emojiional		
	SVM	RF	XGB	SVM	RF	XGB
Accuracy	.56	.53	.51	.57	.54	.56
Macro F1	.51*	.49*	.47*	.53*	.51*	.51*
F1 - Pos	.64	.59	.58	.64	.61	.63
F1 - Neu	.44*	.45*	.41*	.46*	.44*	.46*
F1 - Neg	.44*	.43*	.43*	.48*	.49*	.44*

Note. * indicates improvement from imbalanced datasets.

Overall, all three classifiers achieved comparable accuracy, with scores ranging from 0.51 to 0.57. Models trained with emojiional consistently outperformed those using emoji2vec, with the highest accuracy observed for SVM (0.57). A similar pattern was observed for macro F1, where emojiional yielded improved performance across all classifiers (0.53 for SVM and XGBoost, 0.51 for Random Forest), compared to emoji2vec (0.51, 0.49, and 0.47, respectively).

Class-specific F1 scores further highlight the advantage of emojiional. For the positive class, performance improved modestly across all models, with Random Forest and XGBoost showing the largest gains. The neutral class remains the most challenging, with lower F1 scores overall; however, emojiional still provided consistent improvements. For the negative class, emojiional again outperformed emoji2vec, particularly for Random Forest and XGBoost.

Notably, compared to results on imbalanced datasets, overall accuracy and F1 scores for the positive class declined, while macro F1 scores for the neutral and negative classes improved. This pattern suggests that the models achieved more balanced performance across classes when trained on balanced data. Specifically, the models shifted their focus away from the positive class and better captured patterns in the neutral and negative classes. The previously higher accuracy can thus be

attributed to a bias toward the positive class in the imbalanced setting.

Statistical comparison of model performance was conducted using bootstrapping analysis. Results for accuracy and macro F1 are shown in Figure 4 and Figure 5. Bootstrapping analysis did not find statistical significance across classifiers, nor between emoji2vec and emojiional. In comparison to the imbalanced setting (Figures 1 and 2), several notable changes emerged. While accuracy slightly decreased (but still outperformed baseline), macro F1 scores increased across all classifiers and embeddings, particularly for Random Forest. This shift reflects improved performance on underrepresented classes. Additionally, confidence intervals for macro F1 become more centered and less skewed, suggesting more stable and balanced model behavior.

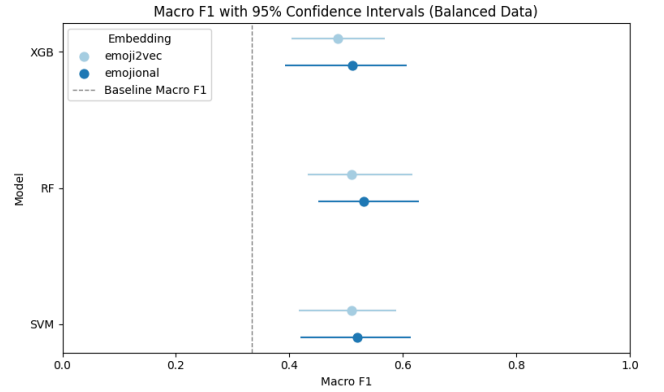


Figure 4. Bootstrapped accuracy of SVM, Random Forest, and XGBoost, across emoji2vec and emojiional using balanced datasets.

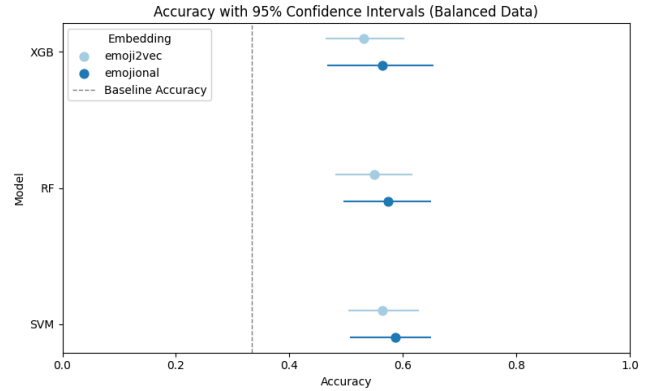


Figure 5. Bootstrapped macro F1 of SVM, Random Forest, and XGBoost, across emoji2vec and emojiional using balanced datasets.

To conclude, when classifiers were trained on imbalanced datasets, higher accuracy was achieved alongside lower macro F1, indicating a bias toward the majority (positive) class. In contrast, the balanced datasets reduced this bias. In particular, models distributed predictive capacity more evenly across classes, leading to improved macro F1 despite minor trade-offs in accuracy. Overall, these results demonstrate that balancing the dataset enhances robustness of the classifiers. However, we did

not find statistically significant differences between emoji2vec and emojiional.

5.4 Hierarchical Structure of Face Emoji Embeddings

To further investigate the emoji semantic structure encoded within each embedding space, we performed hierarchical clustering on a curated subset of face emoji obtained from emoji2vec and emojiional. Results are shown in Figure 6 and Figure 7.

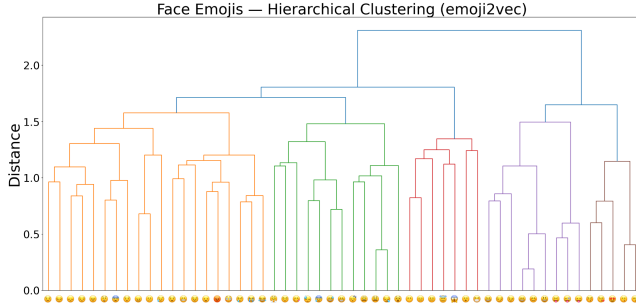


Figure 6. Hierarchical clustering of face emoji using emoji2vec embedding.

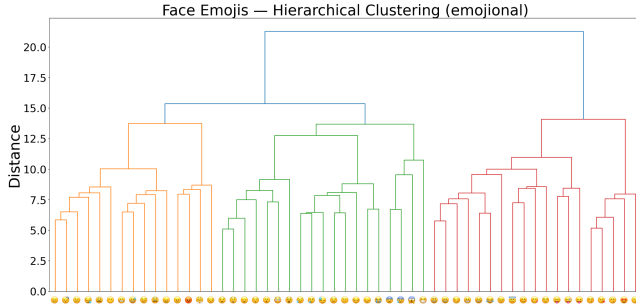


Figure 7. Hierarchical clustering of face emoji using emojiional embedding.

Across both embedding spaces, the hierarchical clustering showed a separation between positively and negatively valenced emoji. This suggests that both models encode emotional valence as a property of their embedding spaces. However, the two graphs notably differed in granularity and structure.

Emoji2vec exhibited more finely differentiated subclusters, suggesting its embeddings space captured more nuanced sentiments. In contrast, emojiional showed broader separation between valence categories. This finding is unexpected, as the researchers who created emojiional aimed to capture a wider range of emotions than emoji2vec, which focused on simple positive vs. negative judgments.

This unexpected finding may be attributed to the fact that the definitions of each emoji in emoji2vec came from restricted official emoji Unicode descriptions, which may contain more of the actual meanings of emoji, compared to emojiional where definitions of emoji were extracted from unrestricted online sites (i.e., Emojipedia and emoji.wiki). These sites may introduce complexity of contexts, leading the model to use broader representations to encompass as many contexts as possible in

each emoji. This difference in sources of emoji definitions may also explain our finding that emojiional revealed a much deeper representational hierarchy compared to emoji2vec. Deeper representational structure may reflect the complexity of semantics and contexts in which emoji are used.

6 Discussion

This study examined whether emoji embedding models capture human judgements of emotional valence by training machine learning classifiers on emoji2vec and emojiional representations. Overall, the models were able to predict emoji valence above baseline performance, indicating that both embedding spaces encode meaningful semantic information. However, performance remained moderate, suggesting that emoji semantics are more complex and context-dependent than traditional text.

With respect to **Hypothesis 1**, the results provide clear support. On balanced datasets, all classifiers achieved above baseline performance, particularly in macro F1. While models did not reach near-perfect accuracies, results were in line with machine learning predictions of word EV [10,11].

In contrast, our findings did not support **Hypothesis 2**. While prior work on word-based sentiment classification has reported higher performance in classifying sentiments, the classifiers in this study only achieved moderate performance, suggesting that emoji are inherently more ambiguous and context dependent than words.

Finally, our results did not support **Hypothesis 3**. Although emojiional consistently achieved slightly higher accuracy and macro F1 than emoji2vec across most classifiers, these differences were not statistically significant under bootstrapping analysis. This suggests that both embedding spaces perform comparably in capturing emotional valence.

6.1 Embedding Space and Human Ratings

Across both embedding spaces, models were able to extract signals related to emotional valence. This suggests that emoji embeddings do encode aspects of human perceived sentiment.

While on imbalanced datasets, models only achieved moderate overall accuracy and uneven class performance in F1 Scores. Class-specific F1 scores on the imbalance datasets were uneven, with all models having relatively strong performance on the positive class, but having substantially lower F1 scores for neutral and negative classes, indicating that these representations are not fully aligned with human judgements.

One consistent pattern observed was the strong performance of SVM relative to Random Forest and XGBoost. This may be attributed to the high-dimensional nature of the embedding space, where SVMs are well-suited to finding separating boundaries between classes. In contrast, Random Forest performed poorly with the imbalance dataset and struggled most to identify minority classes. XGBoost, despite its theoretical advantages for handling imbalance, did not perform as expected, likely due to the relatively small dataset limiting its ability to learn effecting boosting patterns.

Following class balancing with a combination of SMOTE and downsampling, model performance became more evenly distributed across classes. In particular, Random Forest showed

notable improvements compared to its performance on the imbalanced dataset including improved macro F1 and class specific F1 scores for both neutral and negative cases. Additionally, the slight decrease in overall accuracy after balancing suggests that earlier results were influenced by a bias towards the majority class, and that the balanced setting provided a more reliable evaluation of model performance across sentiment categories.

A notable finding is the overall difficulty models faced in predicting neutral emoji. While performance on the positive class was consistently high, neutral and negative classes showed lower F1 scores. This suggests that neutral emoji are inherently more difficult to model, potentially due to their ambiguity and dependence on contextual interpretation.

While emoji2vec showed slightly better overall performance than emoji2vec, the differences between the embeddings were not significant. This suggests that both embeddings capture similar levels of emotional information, despite differences in their training methods.

6.2 Limitations and Future Directions

While the results of this study provide insight into the relationship between emoji embeddings and human judgements of emotional valence, several limitations should be considered when interpreting the findings. Some possible future directions are also outlined.

Temporal Evolution of Emoji Semantics: Emoji semantics are not static, but instead evolve over time. The datasets used in this study reflect specific historical snapshots, including the Emoji Sentiment Ranking dataset (2015) [9], emoji2vec (2016) [5], and emoji2vec (2021) [8]. As a result, both the embeddings and the human annotations are tied to the linguistic and cultural contexts of those periods. However, like words, emoji meaning can shift substantially over time. For example, the “slight smile” emoji was historically associated with a straightforward positive meaning. However with the introduction of a wider range of emoji that more explicitly convey happiness, it is likely the relative simplicity of the original has led to a shift in how it is used. In contemporary contexts, it is frequently used in an ironic manner, often to soften criticism or signal restrained emotion. Consequently, the models evaluated in this study may reflect earlier interpretations of emoji, and may not capture contemporary usage.

Linguistic and Cultural Bias: The embeddings and evaluation data used in this study were derived primarily from English-language sources. Both emoji2vec and emoji2vec rely on English text descriptions. As a result, the learned representations reflect patterns of emoji usage from a specific linguistic and cultural context. However, emoji usage and interpretation can vary across languages and cultures. This introduces a potential bias in the models, as they may implicitly encode English-centric usage patterns. As such, our results may not generalize to other languages and cultures. Future work could extend this analysis by incorporating datasets that sample from a more diverse set of languages and cultural sources to better capture global variation in emoji semantics.

Mixed and Ambiguous Emotional Semantics: Some emoji inherently convey mixed or ambiguous emotional meanings,

which introduces limitations for a classification based approach. While this study categorizes emoji into discrete classes (positive, neutral, and negative), many emoji do not map cleanly onto a single category. Instead, their interpretation often depends heavily on context, tone, and surrounding text. As a result, the use of a fixed three-class labelling scheme may oversimplify the underlying emotional complexity of emoji usage, possibly impacting model performance, especially when attempting to distinguish between neutral and polarized classes.

Dataset Representation and Context Mismatch: The Emoji Sentiment Ranking dataset is only derived from Twitter Data, which may overrepresent high frequency emoji while underrepresenting less commonly used or more context specific ones. This frequency bias can influence both the learned embeddings and the evaluation process, as models are trained and tested on a subset of emoji that may not reflect the full diversity of emoji usage.

Lack of Feature Engineering: While the vector representations of emoji preserve a certain level of semantic meanings, they do not explicitly expose higher level semantic features that may be relevant for distinguishing between sentiment classes. Features related to sentiment intensity, emotional categories, or contextual usage patterns could provide more interpretable signals. Without these intermediate representations, relevant information may remain embedded in the high dimensional space and not be fully utilized by the classifiers. Incorporating feature engineering techniques could therefore improve both model performance and interpretability, particularly in more ambiguous cases such as neutral emoji.

Class Imbalance and Label Ambiguity: The original Emoji Sentiment Ranking dataset contains a substantially higher number of positive emoji compared to neutral and negative ones. Although resampling techniques such as SMOTE and downsampling were applied to address this imbalance, these methods introduce their own constraints. For instance, synthetic data may not fully capture the true distribution of emoji usage, and downsampling may discard useful information about the majority class.

Model Optimization Constraints: While XGBoost was used as a boosting method, more advanced training strategies such as early stopping were not explored. Early stopping halts training when performance on a validation set no longer improves, which could potentially prevent overfitting and improve generalization.

6.3 Applications and Significance

The findings of this study have broader implications for both linguistic research and real-world NLP applications. By examining the extent to which emoji embeddings align with human judgements of emotional valence, this work contributes to understanding how computational models capture evolving forms of digital communication.

Outside linguistics, the results of this study have direct application to real-world NLP systems that rely on accurate interpretation of emoji meaning. Many systems, such as sentiment analysis tools, social media monitoring systems, and content moderation platforms, depend on correctly identifying the emotional tone of user-generated content. Since emoji often carry nuanced meaning, misinterpretation can lead to incorrect

classification of user intent. For example, in a customer support system, accurately understanding positive and negative feedback can help prioritize responses to improve customer experience. Similarly, in a content moderation system, being able to understand the nuanced usage of emoji can assist in identifying harmful, sarcastic, or misleading content that may not be captured through text alone.

The findings of our study are also relevant for the future of large language model (LLM) based systems, like chat bots and AI agents. These systems rely on accurately interpreting user tone in order to generate appropriate and contextually aware responses. However, as shown in this study, emoji meaning is both context-dependent and temporally variable. This presents a challenge for models trained on static datasets that reflect usage from a specific time period. If a LLM is trained on data that does not reflect current emoji usage, it may misinterpret user intent, likely in cases where emoji carry ambiguous or mixed meanings. This could lead the LLM to respond with messages misaligned with the users intentions. As such, ensuring that embedding representations remain aligned with contemporary usage is important for maintaining the reliability of these systems.

7 Conclusion

This study contributes to the growing body of work examining the relationship between computational representations and human interpretation of language by providing a systematic evaluation of emoji embeddings against large scale human-annotated sentiment data. By comparing multiple embedding models (emoji2vec and emojiional) and classification approaches across both imbalanced and balanced datasets, this work highlights how model choice, data distribution, and embedding design influence alignment with human judgements. Our results indicate that emoji embeddings do encode meaningful semantic signals; however, they do not fully reflect the complexity of human interpretation, particularly in cases involving ambiguity or context-dependent meaning. These findings highlight both the potential and the limitations of current embedding-based approaches for modelling emoji semantics.

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